

Linux System Administration & Troubleshooting Lab

Author:

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Role Focus:

IT Support • Help Desk • Desktop Support • Windows System Administration

Lab Objectives

- Gain hands-on experience with the Linux command-line interface.
- Navigate and manage the Linux file system efficiently.
- Create, modify, copy, move, and remove files and directories.
- Configure file ownership and permissions using Linux security commands.
- Monitor system processes and perform basic troubleshooting.
- Execute common networking and diagnostic commands.
- Understand essential Linux administration tasks performed by IT Support, System Administrators, and Cloud professionals.

Component	Details
Host Operating System	Windows 11 Pro
Guest Operating System	Ubuntu Desktop (LTS)
Virtualisation Platform	Oracle VirtualBox
Environment	Home Lab
Tools Used	Terminal, Nano, File System Utilities, Process Management Commands, Network Utilities
Platform	Desktop

Introduction

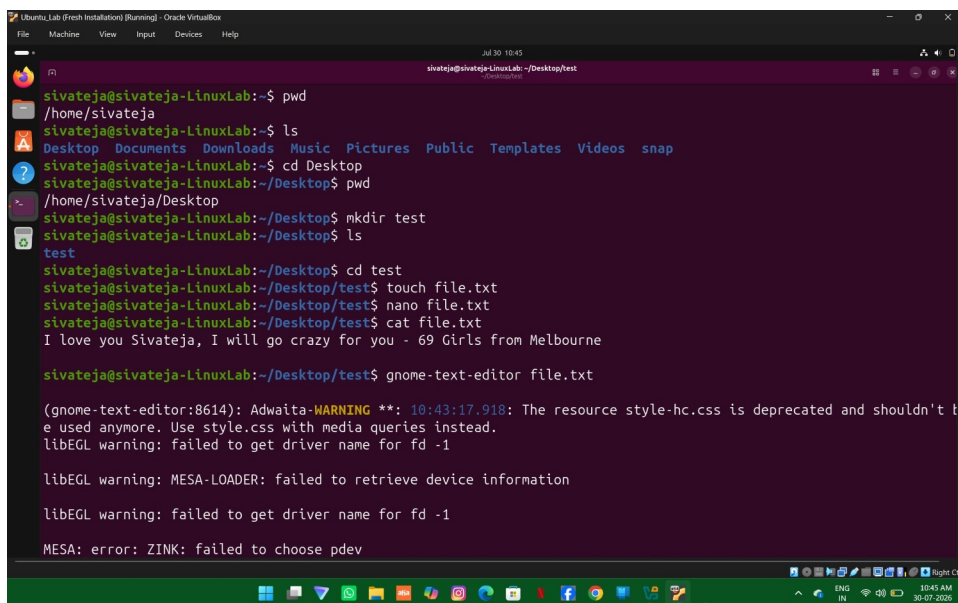
Linux system administration is an essential skill for IT Support, Help Desk, Cloud, and System Administration roles. This hands-on lab demonstrates core Linux administration and troubleshooting tasks using Ubuntu. The exercises cover file system navigation, file management, permissions, process management, networking, and command-line diagnostics.

Basic Linux Commands & File Operations

Practiced fundamental Linux command-line operations by navigating the file system, creating directories and files, editing text files, and viewing file contents. These commands form the foundation of Linux system administration and are used daily for file management, troubleshooting and automation.

Commands Demonstrated

- `pwd` – Display current working directory.
- `ls` – List files and directories.
- `cd` – Change directory.
- `mkdir` – Create a new directory.
- `touch` – Create an empty file.
- `nano` – Edit a text file using the Nano editor.
- `cat` – Display file contents.
- `cat -n` – Display file contents with line numbers.
- `gnome-text-editor` – Open a file using the graphical text editor.



```
sivateja@sivateja-LinuxLab:~$ pwd
/home/sivateja
sivateja@sivateja-LinuxLab:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap
sivateja@sivateja-LinuxLab:~$ cd Desktop
sivateja@sivateja-LinuxLab:~/Desktop$ pwd
/home/sivateja/Desktop
sivateja@sivateja-LinuxLab:~/Desktop$ mkdir test
sivateja@sivateja-LinuxLab:~/Desktop$ ls
test
sivateja@sivateja-LinuxLab:~/Desktop$ cd test
sivateja@sivateja-LinuxLab:~/Desktop/test$ touch file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ nano file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ cat file.txt
I love you Sivateja, I will go crazy for you - 69 Girls from Melbourne

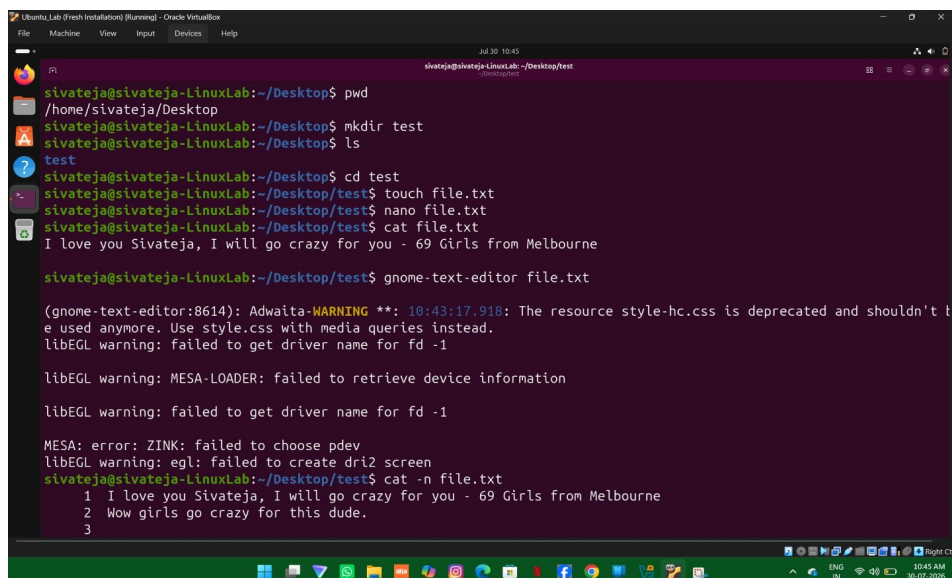
sivateja@sivateja-LinuxLab:~/Desktop/test$ gnome-text-editor file.txt

(gnome-text-editor:8614): Adwaita-WARNING **: 10:43:17.918: The resource style-hc.css is deprecated and shouldn't be
used anymore. Use style.css with media queries instead.
libEGL warning: failed to get driver name for fd -1

libEGL warning: MESA-LOADER: failed to retrieve device information

libEGL warning: failed to get driver name for fd -1

MESA: error: ZINK: failed to choose pdev
```



```
sivateja@sivateja-LinuxLab:~/Desktop$ pwd
/home/sivateja/Desktop
sivateja@sivateja-LinuxLab:~/Desktop$ mkdir test
sivateja@sivateja-LinuxLab:~/Desktop$ ls
test
sivateja@sivateja-LinuxLab:~/Desktop$ cd test
sivateja@sivateja-LinuxLab:~/Desktop/test$ touch file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ nano file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ cat file.txt
I love you Sivateja, I will go crazy for you - 69 Girls from Melbourne

sivateja@sivateja-LinuxLab:~/Desktop/test$ gnome-text-editor file.txt

(gnome-text-editor:8614): Adwaita-WARNING **: 10:43:17.918: The resource style-hc.css is deprecated and shouldn't be
used anymore. Use style.css with media queries instead.
libEGL warning: failed to get driver name for fd -1

libEGL warning: MESA-LOADER: failed to retrieve device information

libEGL warning: failed to get driver name for fd -1

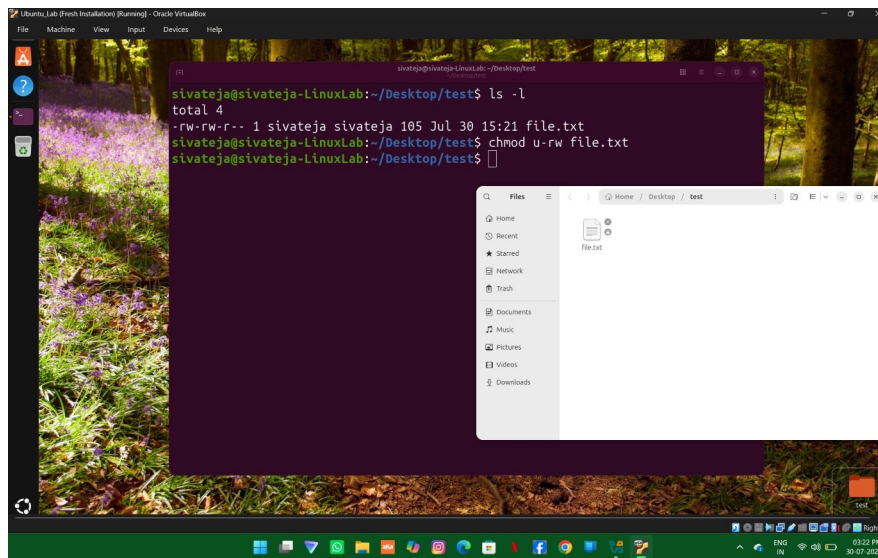
MESA: error: ZINK: failed to choose pdev
libEGL warning: egl: failed to create dri2 screen
sivateja@sivateja-LinuxLab:~/Desktop/test$ cat -n file.txt
 1 I love you Sivateja, I will go crazy for you - 69 Girls from Melbourne
 2 Wow girls go crazy for this dude.
 3
```

File Permissions with chmod

Viewed file permissions using `ls -l` and modified write permissions with the `chmod` command. This exercise demonstrates Linux's permission model and how administrators control access to files and directories.

Commands Demonstrated

- `ls -l` – Display detailed file permissions.
- `chmod` – Modify file permissions.

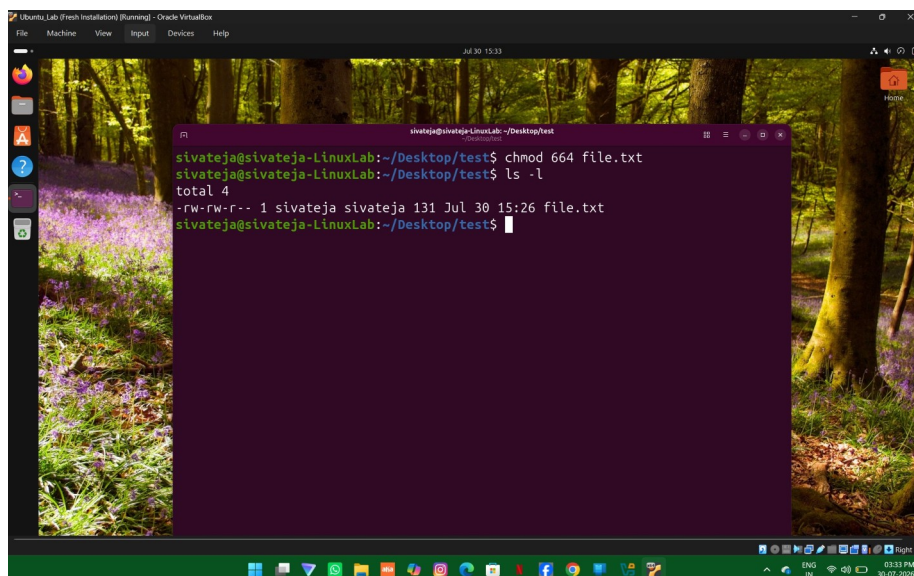


File Permissions with chmod

Viewed file permissions using `ls -l` and modified them using `chmod 664`. This exercise demonstrates Linux's permission model and how administrators control access by assigning different permissions to the file owner, group members, and other users. The value 664 represents Owner = 6 (Read + Write), Group = 6 (Read + Write), Others = 4 (Read Only), where 4 = Read, 2 = Write, and 1 = Execute(x).

Commands Demonstrated

- `ls -l` – Display detailed file permissions.
- `chmod 664` – Set permissions to `rw-rw-r--` (Owner: Read/Write, Group: Read/Write, Others: Read Only).



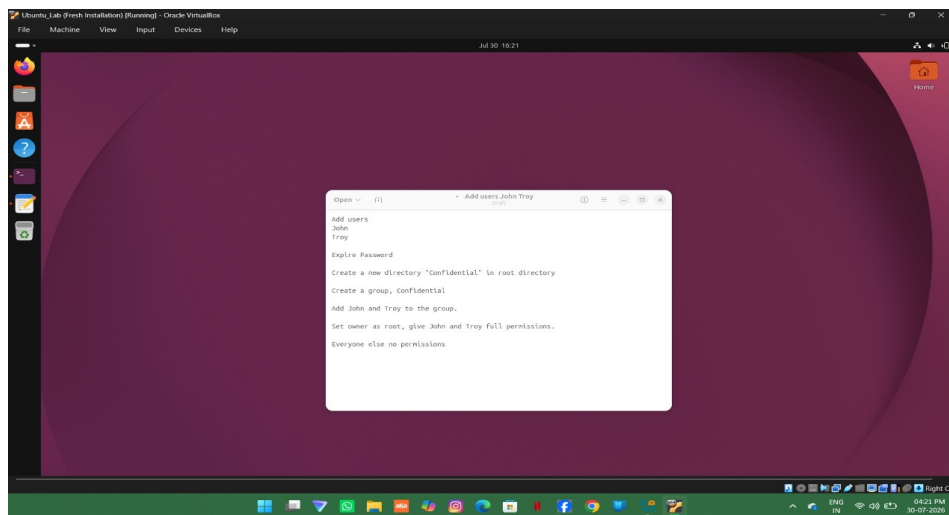
Activity: Secure Access Configuration for the Confidential Department

Scenario

A new Confidential department is being established within the organisation. The IT team has received a request to provision two new users (**John** and **Troy**) and implement secure access controls for confidential company data.

Objectives

- Create local user accounts for John and Troy.
- Create a dedicated security group named confidential.
- Add both users to the confidential group.
- Create a Confidential directory.
- Assign ownership and group permissions appropriately.
- Restrict access so that only authorised members of the confidential group can access the directory.

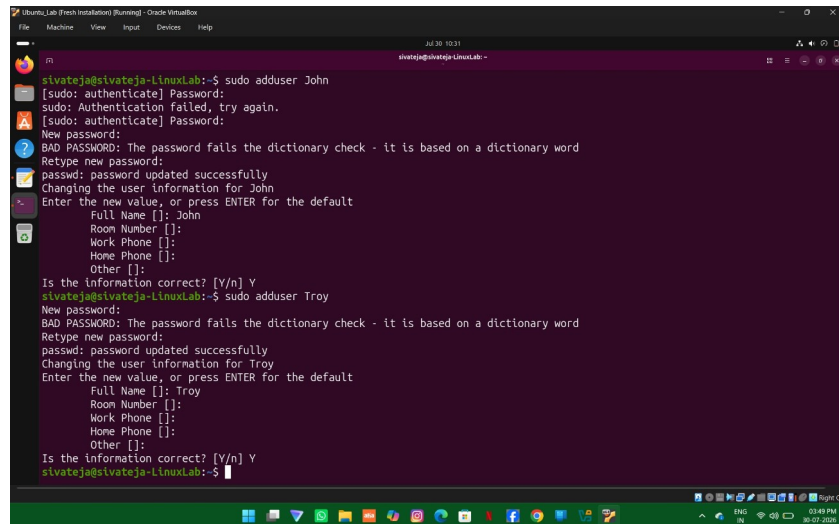


Create Local User Accounts

Created two new local user accounts (John and Troy) using the adduser command. During account creation, passwords and basic user information were configured, provisioning the accounts required for the requested access control implementation.

Commands Demonstrated

- **sudo adduser John** – Create a new local user account named John.
- **sudo adduser Troy** – Create a new local user account named Troy.

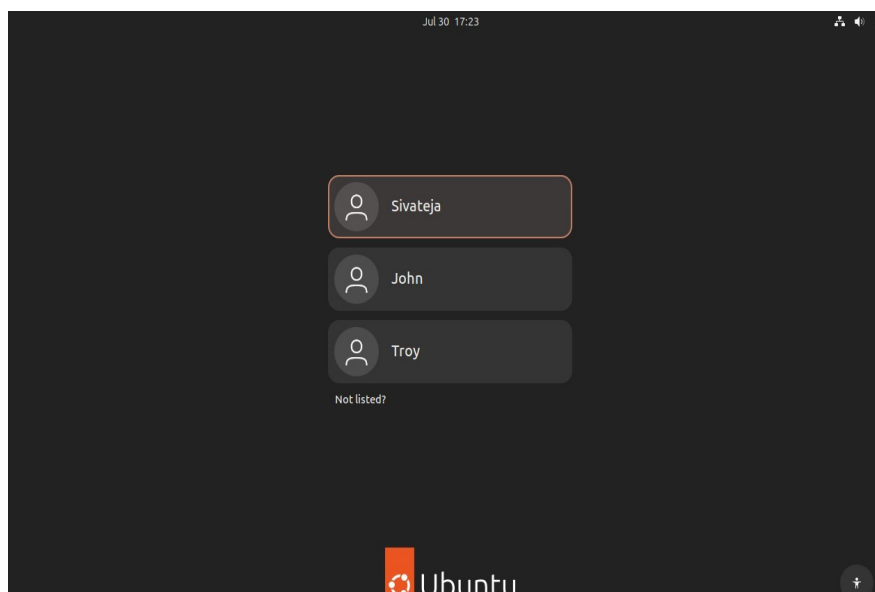


```
sivateja@sivateja-LinuxLab:~$ sudo adduser John
[sudo: authenticate] Password:
[sudo: authentication failed, try again.
[sudo: authenticate] Password:
New password:
BAD PASSWORD: The password falls the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for John
Enter the new value, or press ENTER for the default
  Full Name []: John
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
sivateja@sivateja-LinuxLab:~$ sudo adduser Troy
New password:
BAD PASSWORD: The password falls the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for Troy
Enter the new value, or press ENTER for the default
  Full Name []: Troy
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
sivateja@sivateja-LinuxLab:~$
```

Verify User Account Provisioning

Description

Verified that the newly created user accounts (John and Troy) were successfully provisioned by confirming their availability on the Ubuntu login screen. This ensures the accounts were created correctly and are available for authentication.

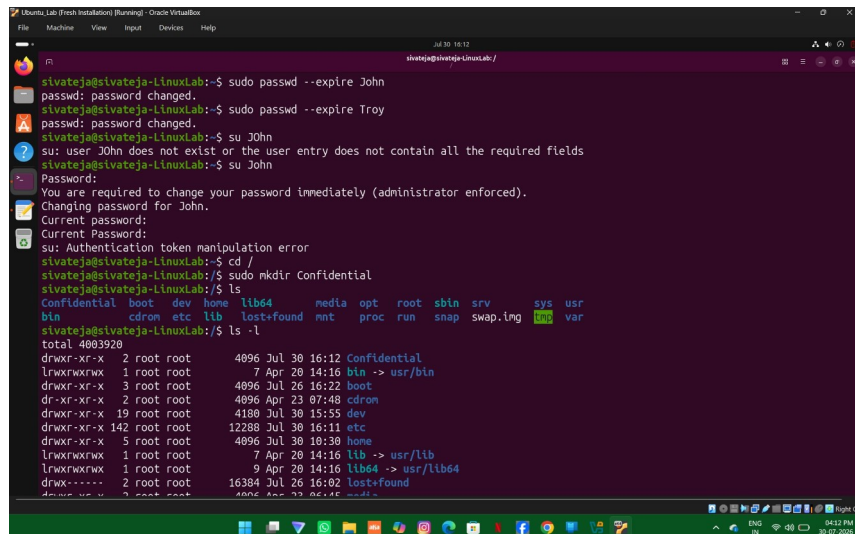


Enforce Password Change at First Login

Configured newly created user accounts to change their passwords during their first login using the `passwd --expire` command. This is a common enterprise security practice that prevents administrators from permanently knowing user passwords.

Commands Demonstrated

- **sudo passwd --expire John** – Force John to change password at next login.
- **sudo passwd --expire Troy** – Force Troy to change password at next login.
- **su John** – Verify the password change policy is enforced.



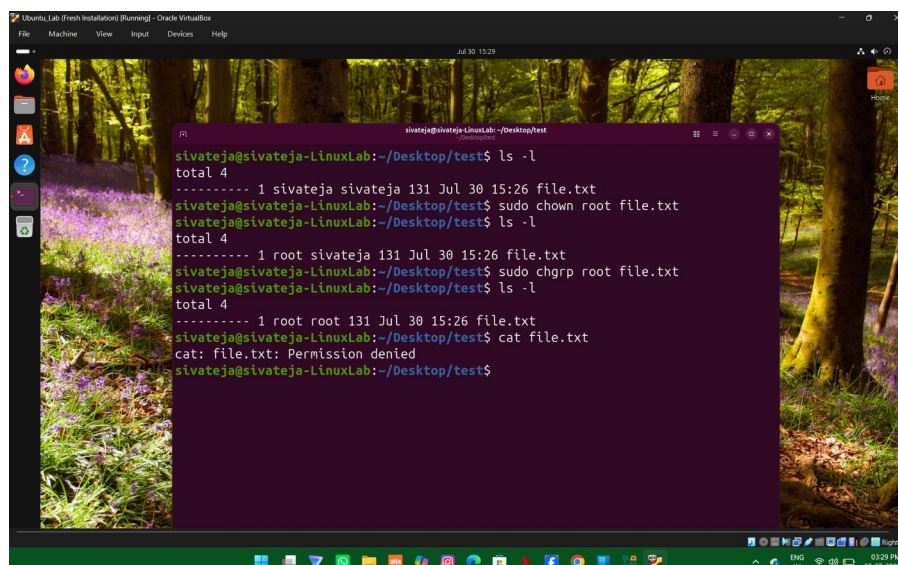
A terminal window in a virtual machine showing the following commands and output:

```
sivateja@sivateja-LinuxLab:~$ sudo passwd --expire John
passwd: password changed.
sivateja@sivateja-LinuxLab:~$ sudo passwd --expire Troy
passwd: password changed.
sivateja@sivateja-LinuxLab:~$ su John
su: user John does not exist or the user entry does not contain all the required fields
sivateja@sivateja-LinuxLab:~$ su John
Password:
You are required to change your password immediately (administrator enforced).
Changing password for John.
Current password:
Current Password:
su: Authentication token manipulation error
sivateja@sivateja-LinuxLab:~$ cd /
sivateja@sivateja-LinuxLab:/$ sudo mkdir Confidential
sivateja@sivateja-LinuxLab:/$ ls
Confidential boot dev home lib64 media opt root sbin srv sys usr
bin cdrom etc lib lost+found mnt proc run snap swap.ing var
sivateja@sivateja-LinuxLab:/$ ls -l
total 4003920
drwxr-xr-x 2 root root 4096 Jul 30 16:12 Confidential
lrwxrwxrwx 1 root root 7 Apr 20 14:16 bin -> usr/bin
drwxr-xr-x 3 root root 4096 Jul 26 16:22 boot -
dr-xr-xr-x 2 root root 4096 Apr 23 07:48 cdrom
drwxr-xr-x 19 root root 4188 Jul 30 15:55 dev
drwxr-xr-x 142 root root 12288 Jul 30 16:11 etc
drwxr-xr-x 5 root root 4096 Jul 30 18:30 home
lrwxrwxrwx 1 root root 7 Apr 20 14:16 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Apr 20 14:16 lib64 -> usr/lib64
drwx----- 2 root root 16384 Jul 26 16:02 lost+found
drwxr-xr-x 2 root root 4096 Apr 23 06:46 media
```

Create Secure Shared Directory

Created a shared directory named `Confidential` in the root filesystem to store sensitive company information. This directory serves as the protected resource that will later be secured using Linux ownership, groups, and permissions.

- **cd /** – Navigate to the root directory.
- **sudo mkdir Confidential** – Create the shared directory.
- **ls -l** – Verify the directory was successfully created.



A terminal window in a virtual machine showing the following commands and output:

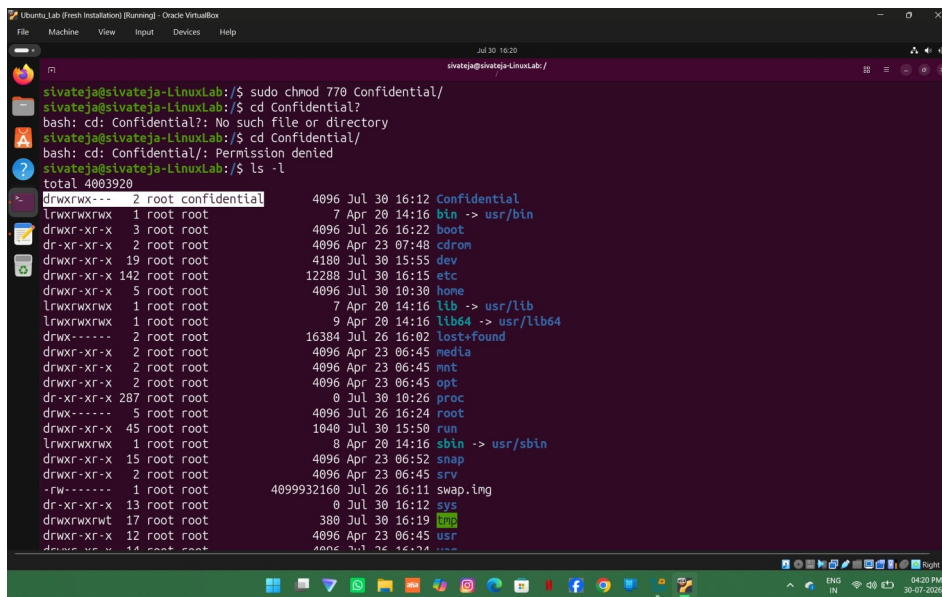
```
sivateja@sivateja-LinuxLab:~/Desktop/test$ ls -l
total 4
----- 1 sivateja sivateja 131 Jul 30 15:26 file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ sudo chown root file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ ls -l
total 4
----- 1 root sivateja 131 Jul 30 15:26 file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ sudo chgrp root file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ ls -l
total 4
----- 1 root root 131 Jul 30 15:26 file.txt
sivateja@sivateja-LinuxLab:~/Desktop/test$ cat file.txt
cat: file.txt: Permission denied
sivateja@sivateja-LinuxLab:~/Desktop/test$
```

Configure Ownership, Groups and Access Permissions

Assigned ownership of the Confidential directory to the root user and the confidential group, then applied Linux permissions to restrict access. Permissions were configured using `chmod 770`, allowing only the owner and authorised group members full access while denying access to all other users. Verification confirmed that unauthorised users were unable to access the protected directory.

Commands Demonstrated

- **`sudo chown root:confidential Confidential`** – Assign owner and group.
- **`sudo chmod 770 Confidential`** – Grant full access to owner and group only.
- **`ls -l`** – Verify ownership and permissions.
- **`cd Confidential`** – Verify unauthorised users are denied access.



```
Ubuntu Lab (Fresh Installation) (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help

sivateja@sivateja-LinuxLab: /
sivateja@sivateja-LinuxLab:/$ sudo chown root:confidential Confidential/
sivateja@sivateja-LinuxLab:/$ cd Confidential/
bash: cd: Confidential?: No such file or directory
sivateja@sivateja-LinuxLab:/$ cd Confidential/
bash: cd: Confidential/: Permission denied
sivateja@sivateja-LinuxLab:/$ ls -l
total 4083920
drwxrwx--- 2 root confidential 4096 Jul 30 16:12 Confidential
lrwxrwxrwx 1 root root 7 Apr 20 14:16 bin -> usr/bin
drwxr-xr-x 3 root root 4096 Jul 26 16:22 boot
dr-xr-xr-x 2 root root 4096 Apr 23 07:48 cdrom
drwxr-xr-x 19 root root 4188 Jul 30 15:55 dev
drwxr-xr-x 142 root root 12288 Jul 30 16:15 etc
drwxr-xr-x 5 root root 4096 Jul 30 18:30 home
lrwxrwxrwx 1 root root 7 Apr 20 14:16 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Apr 20 14:16 lib64 -> usr/lib64
drwx----- 2 root root 16384 Jul 26 16:02 lost+found
drwxr-xr-x 2 root root 4096 Apr 23 06:45 media
drwxr-xr-x 2 root root 4096 Apr 23 06:45 mnt
drwxr-xr-x 2 root root 4096 Apr 23 06:45 opt
dr-xr-xr-x 287 root root 0 Jul 30 18:26 proc
drwx----- 5 root root 4096 Jul 26 16:24 root
drwxr-xr-x 45 root root 1048 Jul 30 15:58 run
lrwxrwxrwx 1 root root 8 Apr 20 14:16/sbin -> usr/sbin
drwxr-xr-x 15 root root 4096 Apr 23 06:52 snap
drwxr-xr-x 2 root root 4096 Apr 23 06:45 srv
-rw----- 1 root root 4099932160 Jul 26 16:11 swap.img
dr-xr-xr-x 13 root root 0 Jul 30 16:12 sys
drwxrwxrwt 17 root root 388 Jul 30 16:19 tmp
drwxr-xr-x 12 root root 4096 Apr 23 06:45 usr
drwxr-xr-x 14 root root 4096 Jul 26 16:24 uss
```

Package Management (APT)

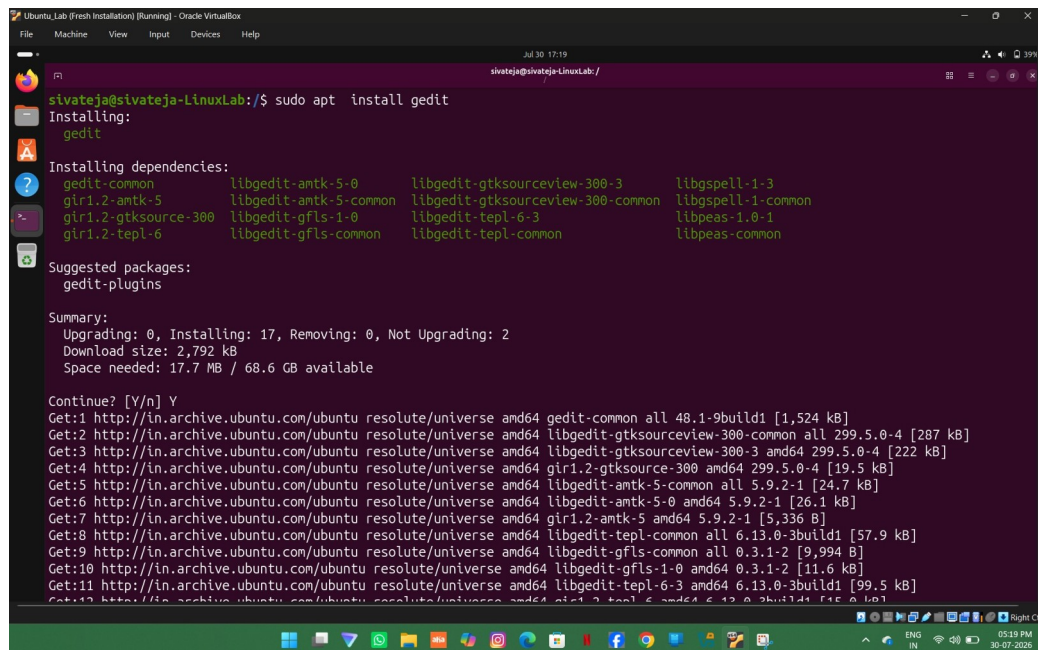
Installed and removed software packages using Ubuntu's APT (Advanced Package Tool) package manager. This demonstrates how Linux administrators install applications, resolve dependencies, and remove unnecessary software from a system.

Commands Demonstrated

- **`sudo apt install gedit`** – Install the gedit text editor and required dependencies.
- **`sudo apt remove gedit`** – Remove the installed package while retaining configuration files.

Installing Packages

Installed the gedit text editor using the APT package manager. During installation, APT automatically downloaded and installed all required dependency packages before configuring the application.



```
Ubuntu Lab (Fresh Installation) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 30 17:19
sivateja@sivateja-LinuxLab: /
sivateja@sivateja-LinuxLab:/$ sudo apt install gedit
Installing:
  gedit

Installing dependencies:
  gedit-common      libgedit-amtk-5-0      libgedit-gtksourceview-300-3      libgspell-1-3
  gir1.2-amtk-5      libgedit-amtk-5-common  libgedit-gtksourceview-300-common  libgspell-1-common
  gir1.2-gtksource-300  libgedit-gfls-1-0      libgedit-tepl-6-3                  libpeas-1.0-1
  gir1.2-tepl-6        libgedit-gfls-common    libgedit-tepl-common                libpeas-common

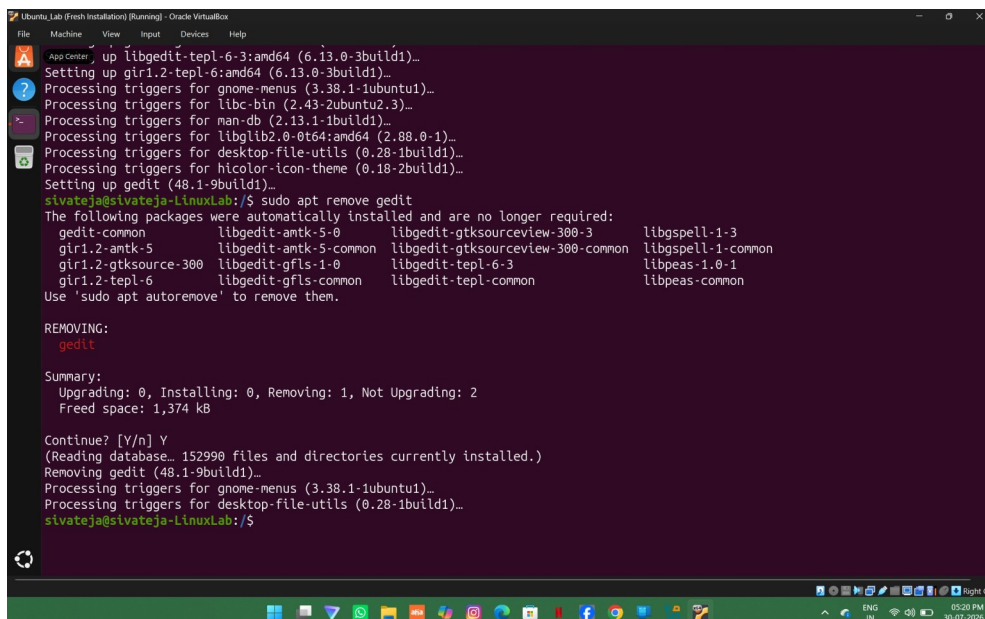
Suggested packages:
  gedit-plugins

Summary:
  Upgrading: 0, Installing: 17, Removing: 0, Not Upgrading: 2
  Download size: 2,792 kB
  Space needed: 17.7 MB / 68.6 GB available

Continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 gedit-common all 48.1-9build1 [1,524 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-gtksourceview-300-common all 299.5.0-4 [287 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-gtksourceview-300-3 amd64 299.5.0-4 [222 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 gir1.2-gtksource-300 amd64 299.5.0-4 [19.5 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-amtk-5-common all 5.9.2-1 [24.7 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-amtk-5-0 amd64 5.9.2-1 [26.1 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 gir1.2-amtk-5 amd64 5.9.2-1 [5,336 B]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-tepl-common all 6.13.0-3build1 [57.9 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-gfls-common all 0.3.1-2 [9,994 B]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-gfls-1-0 amd64 0.3.1-2 [11.6 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libgedit-tepl-6-3 amd64 6.13.0-3build1 [99.5 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 gir1.2-tepl-6 amd64 6.13.0-3build1 [14.0 kB]
```

Removing Packages

Removed the gedit package using APT. The package was successfully uninstalled while unused dependencies were identified for optional removal using sudo apt autoremove. This demonstrates basic software lifecycle management on Ubuntu.



```
Ubuntu Lab (Fresh Installation) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
App Center, up libgedit-tepl-6-3:amd64 (6.13.0-3build1)...
Setting up gir1.2-tepl-6:amd64 (6.13.0-3build1)...
Processing triggers for gnome-menus (3.38.1-1ubuntu1)...
Processing triggers for libc-bin (2.43-2ubuntu2.3)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libgl1:amd64 (2.80.0-1)...
Processing triggers for desktop-file-utils (0.28-1build1)...
Processing triggers for hicolor-icon-theme (0.18-2build1)...
Setting up gedit (48.1-9build1)...
sivateja@sivateja-LinuxLab:/$ sudo apt remove gedit
The following packages were automatically installed and are no longer required:
  gedit-common      libgedit-amtk-5-0      libgedit-gtksourceview-300-3      libgspell-1-3
  gir1.2-amtk-5      libgedit-amtk-5-common  libgedit-gtksourceview-300-common  libgspell-1-common
  gir1.2-gtksource-300  libgedit-gfls-1-0      libgedit-tepl-6-3                  libpeas-1.0-1
  gir1.2-tepl-6        libgedit-gfls-common    libgedit-tepl-common                libpeas-common
Use 'sudo apt autoremove' to remove them.

REMOVING:
  gedit

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 2
  Freed space: 1,374 kB

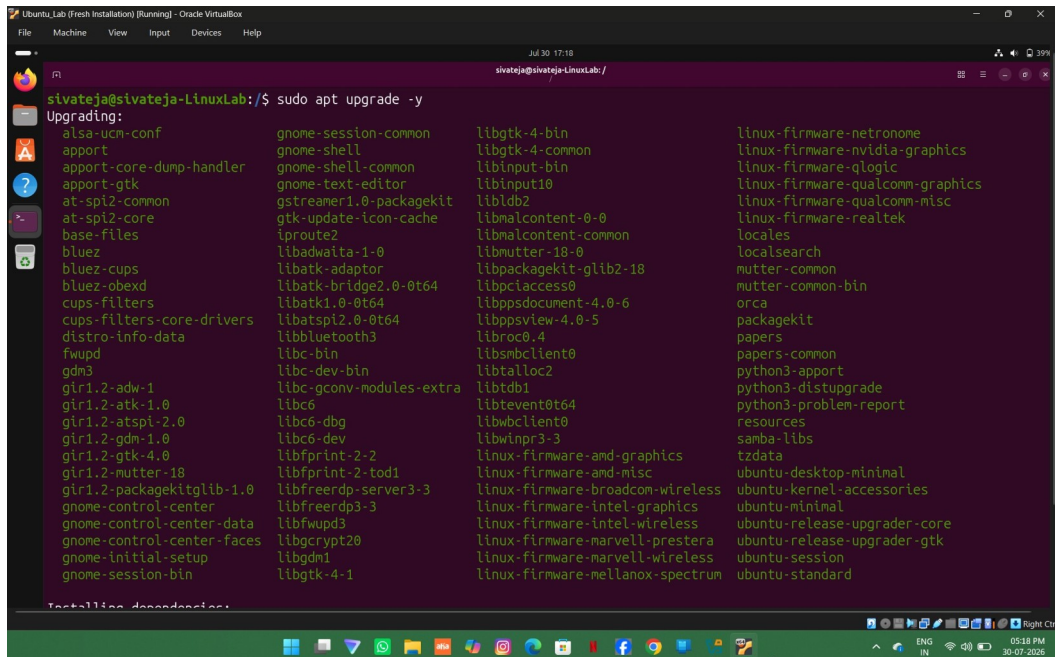
Continue? [Y/n] Y
(Reading database... 152990 files and directories currently installed.)
Removing gedit (48.1-9build1)...
Processing triggers for gnome-menus (3.38.1-1ubuntu1)...
Processing triggers for desktop-file-utils (0.28-1build1)...
sivateja@sivateja-LinuxLab:/$
```


System Package Upgrade

Updated all installed packages to their latest available versions using the APT package manager. This demonstrates routine Linux system maintenance by downloading and installing security patches, bug fixes, and software updates.

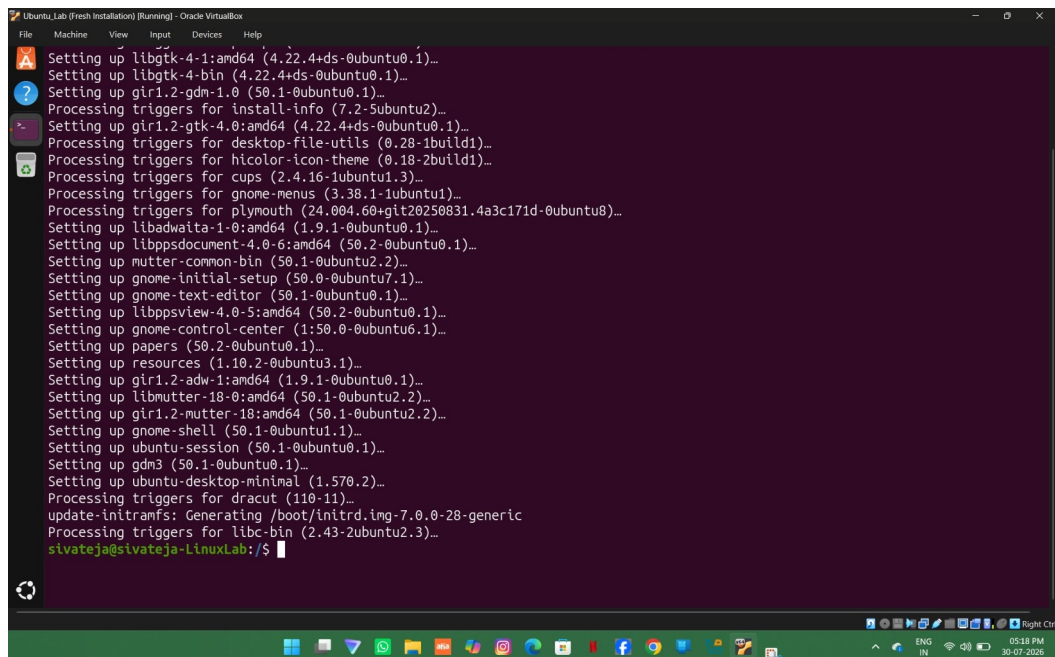
Commands Demonstrated

- **sudo apt upgrade -y** – Upgrade all installed packages and automatically confirm the installation.



```
Ubuntu Lab (Fresh Installation) [Running] - Oracle VM VirtualBox
Jul 30 17:18
sivateja@sivateja-LinuxLab: /$ sudo apt upgrade -y
Upgrading:
alsacv-conf      gnome-session-common  libgtk-4-bin          linux-firmware-netronome
apport           gnome-shell           libgtk-4-common       linux-firmware-nvidia-graphics
apport-core-dump-handler  gnome-shell-common    libinput-bin          linux-firmware-qlogic
apport-gtk       gnome-text-editor     libinput10            linux-firmware-qualcomm-graphics
at-spi2-common   gstreamer1.0-packagekit  libldb2              linux-firmware-qualcomm-misc
at-spi2-core     gtk-update-icon-cache  libmalcontent-0-0     linux-firmware-realtek
base-files       iproute2              libmalcontent-common  locales
bluez            libadwaita-1-0        libmutter-18-0        localsearch
bluez-cups       libatk-adaptor        libpackagekit-glib2-18  mutter-common
bluez-obexd      libatk-bridge2.0-0t64  libpciaccess0         mutter-common-bin
cups-filters     libatk1.0-0t64        libppsd-4.0-6         orca
cups-filters-core-drivers  libatspi2.0-0t64     libppsv-4.0-5         packagekit
distro-info-data  libbluetooth3         libroc0.4             papers
fwupd            libbpf-bin            libsbml-client0       papers-common
gdm3             libc-dev-bin          libtalloc2            python3-apport
gir1.2-adw-1      libc-gconv-modules-extra  libtdb1              python3-distupgrade
gir1.2-atk-1.0    libc6                 libtvent-0t64         python3-problem-report
gir1.2-at-spi-2.0  libc6-dbg             libwbclient0          resources
gir1.2-gdm-1.0    libc6-dev             libwinpr3-3           samba-ls
gir1.2-gtk-4.0    libfprint-2-2         linux-firmware-amd-graphics  tzdata
gir1.2-mutter-18  libfprint-2-tod1      linux-firmware-amd-misc  ubuntu-desktop-minimal
gir1.2-packagekit-glib-1.0  libfreerdp-server3-3  linux-firmware-broadcom-wireless  ubuntu-kernel-accessories
gnome-control-center  libfreerdp3-3        linux-firmware-intel-graphics  ubuntu-minimal
gnome-control-center-data  libfwpd3             linux-firmware-intel-wireless  ubuntu-release-upgrader-core
gnome-control-center-faces  libgcrpt20           linux-firmware-marvell-prestera  ubuntu-release-upgrader-gtk
gnome-initial-setup  libgdm1              linux-firmware-marvell-wireless  ubuntu-session
gnome-session-bin   libgtk-4-1           linux-firmware-mellanox-spectrum  ubuntu-standard

Installing dependencies:
```



```
Setting up libgtk-4-1:amd64 (4.22.4+ds-0ubuntu0.1)...
Setting up libgtk-4-bin (4.22.4+ds-0ubuntu0.1)...
Setting up gir1.2-gdm-1.0 (50.1-0ubuntu0.1)...
Processing triggers for install-info (7.2-Subuntu2)...
Setting up gir1.2-gtk-4.0:amd64 (4.22.4+ds-0ubuntu0.1)...
Processing triggers for desktop-file-utils (0.28-1build1)...
Processing triggers for hicolor-icon-theme (0.18-2build1)...
Processing triggers for cups (2.4.16-1ubuntu1.3)...
Processing triggers for gnome-menus (3.38.1-1ubuntu1)...
Processing triggers for plymouth (24.004.60+git20250831.4a3c171d-0ubuntu8)...
Setting up libadwaita-1-0:amd64 (1.9.1-0ubuntu0.1)...
Setting up libppsd-4.0-6:amd64 (50.2-0ubuntu0.1)...
Setting up mutter-common-bin (50.1-0ubuntu2.2)...
Setting up gnome-initial-setup (50.0-0ubuntu7.1)...
Setting up gnome-text-editor (50.1-0ubuntu0.1)...
Setting up libppsv-4.0-5:amd64 (50.2-0ubuntu0.1)...
Setting up gnome-control-center (1:50.0-0ubuntu6.1)...
Setting up papers (50.2-0ubuntu0.1)...
Setting up resources (1:10.2-0ubuntu3.1)...
Setting up gir1.2-adw-1:amd64 (1.9.1-0ubuntu0.1)...
Setting up libmutter-18-0:amd64 (50.1-0ubuntu2.2)...
Setting up gir1.2-mutter-18:amd64 (50.1-0ubuntu2.2)...
Setting up gnome-shell (50.1-0ubuntu1.1)...
Setting up ubuntu-session (50.1-0ubuntu0.1)...
Setting up gdm3 (50.1-0ubuntu0.1)...
Setting up ubuntu-desktop-minimal (1.570.2)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-28-generic
Processing triggers for libc-bin (2.43-2ubuntu2.3)...
sivateja@sivateja-LinuxLab:/$
```

System Update Verification

Verified that the Ubuntu system is fully updated after running package update and upgrade operations. The Software Updater confirmed that no additional updates were available, indicating the system is running the latest installed packages.

Commands Demonstrated

- **sudo apt update** – Refresh package information from configured repositories.
- **sudo apt upgrade** – Install available package updates.
- **Software Updater** – Verify that the operating system is fully up to date.

```
stivateja@stivateja-LinuxLab:/$ sudo apt update && sudo apt upgrade
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
gedit-common libgedit-antk-5-0 libgedit-gtksourceview-300-3 libgspell-1-3
gir1.2-antk-5 libgedit-antk-5-common libgedit-gtksourceview-300-common libgspell-1-common
gir1.2-gtksource-300 libgedit-gfls-1-0 libgedit-tepl-6-3 libpeas-1.0-1
gir1.2-tepl-6 libgedit-gfls-common libgedit-tepl-common libpeas-common
Use 'sudo apt autoremove' to remove them.
Not upgrading yet due to phasing:
python3-software-properties software-properties
Summary:
Upgrading: 0, Installing: 0, Removing: 0,
stivateja@stivateja-LinuxLab:/$
```

Conclusion

This lab provided hands-on experience with essential Linux system administration tasks using Ubuntu in a VirtualBox environment. Practical exercises covered Linux command-line fundamentals, file and directory management, user and group administration, file permissions, ownership, package management, and system updates. These activities reflect common responsibilities performed by IT Support, Help Desk, System Administration, and Cloud Operations professionals in real-world enterprise environments. Completing this lab strengthened practical Linux administration skills and reinforced core concepts required for managing, troubleshooting, and maintaining Linux systems.